

- 7 Scientists investigated the numbers and types of mutations in different species of animal.

Samples of tissue were collected from living animals of each species and the types of mutation determined.

- (a) The types of mutation were classed as either substitution mutations or insertion and deletion mutations.

The table shows the results of this investigation.

Species of animal	Number of animals sampled	Range of number of mutations per cell	
		Substitution	Insertion and deletion
Naked mole rat	3	100 to 850	10 to 50
Mouse	10	300 to 2 000	50 to 400
Dog	3	1 000 to 3 000	50 to 300
Human	8	200 to 350	10 to 200

- (i) Describe how these types of mutation affect the **DNA base sequence** of a gene.

(2)

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- (ii) Explain why these animals can survive with a greater number of substitution mutations than insertion and deletion mutations.

(4)

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- (iii) Suggest which set of data is the most valid.

Give a reason for your answer.

(1)

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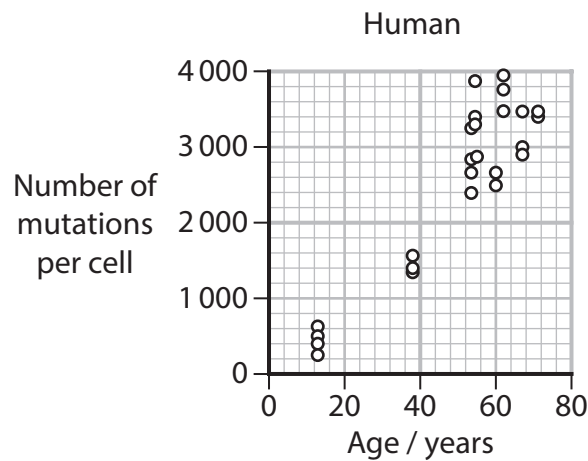
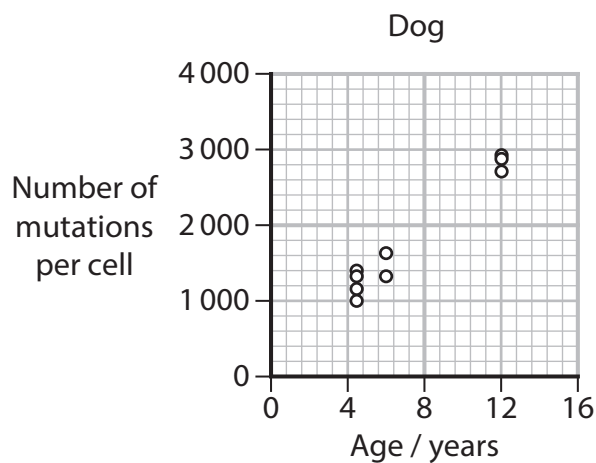
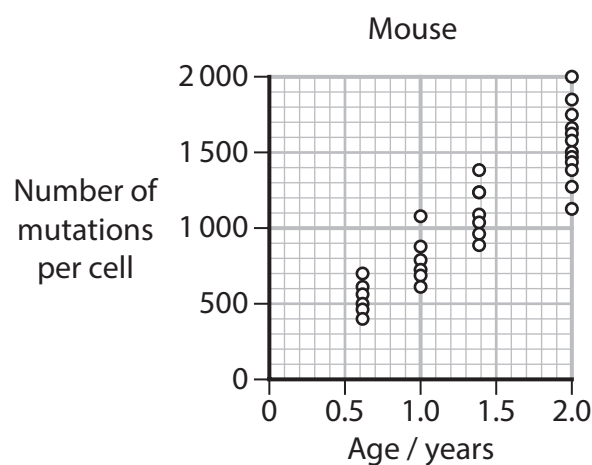
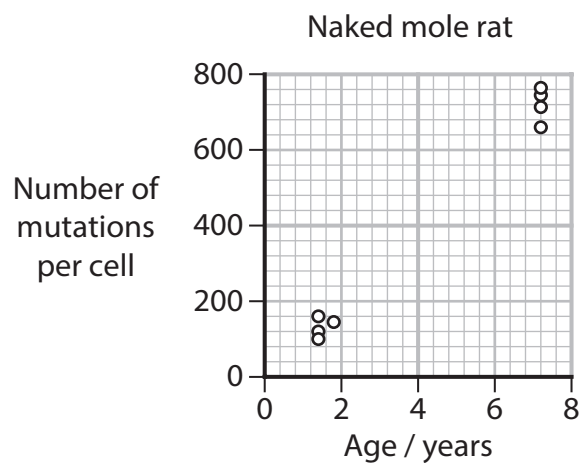
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- (b) The graphs show the number of mutations per cell during the lifespans of these four animals.



- (i) Calculate the mean mutation rate for the mouse.

Draw a line of best fit onto the graph to help you.

(2)

Answer number of mutations per cell year⁻¹



(ii) Describe **three** conclusions that can be made from these graphs.

(3)

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3

(iii) Suggest **two** reasons for the differences in the number of mutations per cell of these species of animal.

(2)

1

2

(Total for Question 7 = 14 marks)

